Yanamala Shamili

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	6.78
Narayana Junior College , Madinaguda	Class XII (Telangana State Board of Intermediate Education)	2023	98.38%
Narayana School KPHB colony	Class X (Board of Secondary Education, Telangana)	2021	10.00%

SKILLS

- **Programming Languages**: Python, C++, C
- **Libraries and Framework**: PyTorch, NumPy, Pandas, Matplotlib, OpenCV, spaCy, Hugging Face Transformers
- **Tools**: Git/GitHub, SolidEdge, PowerBI, Jupyter Notebook, Kaggle

RELEVANT COURSES

- **Computer Science**: Introduction to programming(C, OOP, File handling fundamentals), Software Development Fundamentals(Design principles, Web development basics), Data Structures, Algorithms, Operating Systems(xv6 Kernel Programming), Database Management Systems, Theory of Computation, Computer Architecture(Risc-v)
- Mathematics: Linear Algebra, Discrete Mathematics, Calculus, Probability and Random variables, Statistics

EXPERIENCE

BSNL Networking Intern:

- Studied fundamental networking concepts including IPv4 addressing, subnetting, IP address classes, and network devices.
- Used Cisco packet Tracer to understand basic network configurations and topologies.
- Prepared and delivered a presentation on IPv4 Address Classes(A-E) .

PROJECTS

Arbitrary Arithmetic Precision:

- Technologies: Java, Python, Apache Ant, OOP
- Implemented arbitrary-precision integer and floating point arithmetic from scratch in Java, supporting numbers beyond native Long/double limits using string based digit manipulation.
- Designed AInteger and AFloat classes with **sign-magnitude representation**, implementing addition, subtraction, multiplication, and division via digit-by-digit carry/borrow propagation and grade-school long-division algorithms.
- Diagnosed and fixed a **critical infinite-loop bug** in the floating-point division algorithm caused by incorrect remainder propagation, restoring correct O(n) per-digit division performance.
- Built a **Python-based build/test orchestrator** using Apache Ant to automate compilation and CLI-driven execution of the Java arithmetic engine.

RISC-V Simulator:

- Developed and extended a modular RISC-V CPU simulator in C++, implementing a complete **five-stage instruction pipeline** consisting of **IF, ID, EX, MEM, and WB stages**.
- Designed mechanisms for **data hazard detection and dependency management**, enabling correct execution of dependent instructions through pipeline coordination and stall handling.
- Implemented **data forwarding techniques** to reduce unnecessary pipeline stalls by forwarding intermediate results directly between pipeline stages whenever dependencies occur.

- Integrated **branch prediction and control hazard handling mechanisms** to manage branch instructions and improve instruction flow and overall pipeline efficiency.

Code Comment Autocorrection using T5:

- Technologies: Python, PyTorch, Hugging Face Transformers, Gradio
- Fine-tuned a **T5-small transformer** to automatically detect and correct typographical errors in Python and C++ code comments using surrounding code context.
- Built a synthetic typo-generation pipeline on **15,000 CodeSearchNet samples** to create training data and trained the model using **Kaggle T4 GPUs**.
- Evaluated the model on unseen code comments to assess contextual typo correction performance.
- Developed an interactive **Gradio** web interface for real-time typo correction and model inference.

ISL(Indian Sign language Recognition - Long term Project):

- Technologies: Python, spaCy, OpenCV, NLP
- Built an offline ISL translation application using a modular NLP pipeline for converting text into ISL video sequences.
- Implemented rule based sentence restructuring and POS tagging to translate english grammar into ISL grammar.
- Developed an efficient video indexing and playback system using 1,266 ISL video clips, providing 95.8% vocabulary coverage.
- Architected the system for scalability with support for future speech-to-sign and sign-to-text modules.

CERTIFICATIONS

Quantum Fundamentals Program-WISER, Amaravati Quantum Valley & Qubitech:

- 4-week program covering qubits, quantum gates, superposition and quantum circuits.

AI For Everyone-Andrew Ng, Coursera:

- Covered AI fundamentals, ML concepts, AI project workflows , applications, and societal impact.

EXTRACURRICULAR

- Art: Practicing sketching and exploring visual art, with experience participating in local art competitions.
- Reading: Interested in reading self-improvement and fiction books, with a focus on personal growth.